Интегралы

Семинары 7

1а) $\int 3 \cdot 2^x dx = 3 \cdot \int 2^x dx = 3 \cdot \frac{2^x}{\ln 2} + C$

1б) $\int \left(x^3 - 4x^2\right) dx = \frac{x^4}{4} - \frac{4x^3}{3} + C$

1в) $\int cos\left(2x - 5\right) dx = \frac{1}{2} sin\left(2x - 5\right) + C$

1г) $\int \sqrt{4x + 7}\, dx = \int \left(4x + 7\right)^{\frac{1}{2}} dx = \frac{\left(4x + 7\right)^{\frac{3}{2}}}{\frac{3}{2} \cdot 4} + C = \frac{\sqrt{\left(4x + 7\right)^3}}{6} + C$

1д) $\int \frac{4x + 3}{4x + 1} dx = \int \frac{\left(4x + 1\right) + 2}{4x + 1} dx = \int \left(1 + \frac{2}{4x + 1}\right) dx =$

$$= x + \frac{2\ln\left|4x + 1\right|}{4} + C = x + \frac{\ln\left|4x + 1\right|}{2} + C$$

$$\int_a^b f(x)dx = F(x)\,\Big|_a^b = F(b) - F(a)$$

2а) $\int_{-4}^{-1} \frac{dx}{\sqrt{x}}$ не сущ

2б) $\int_1^4 \frac{dx}{\sqrt{x}} = \int_1^4 x^{-0.5}dx = \frac{x^{0.5}}{0.5}\,\Big|_1^4 = 2\sqrt{x}\,\Big|_1^4 = 2\sqrt{4} - 2\sqrt{1} = 4 - 2 = 2$

2в) $\int_0^{ln2} e^x dx = e^x\,\Big|_0^{ln2} = e^{ln2} - e^0 = 2 - 1 = 1$

2г) $\int_{-\pi}^{\frac{\pi}{2}} sin(3x - \pi)dx = \frac{-cos(3x - \pi)}{3}\,\Big|_{-\pi}^{\frac{\pi}{2}} = \frac{-cos\left(\frac{3\pi}{2} - \pi\right)}{3} - \left(\frac{-cos(-3\pi - \pi)}{3}\right) =$

$$= \frac{-cos\left(\frac{\pi}{2}\right)}{3} - \left(\frac{-cos(-4\pi)}{3}\right) = 0 + \frac{1}{3} = \frac{1}{3}$$

3а) Вычислить площадь фигуры, ограниченной

$$f(x)=\frac{1}{cos^2 x},\ y=0,\ x=-\frac{\pi}{6},\ x=\frac{\pi}{3}$$

$$D(y):\ x\neq\frac{\pi}{2}+\pi k,\ k\in\mathbb{Z},$$

но наш промежуток хороший, не содержит таких точек!

$f(x)>0$ всегда, поэтому

$$S=\int_{-\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{dx}{cos^2 x}=tgx\ \Big|_{-\frac{\pi}{6}}^{\frac{\pi}{3}}=tg\frac{\pi}{3}-tg\left(-\frac{\pi}{6}\right)=tg\frac{\pi}{3}+tg\left(\frac{\pi}{6}\right)=$$

$$=\sqrt{3}+\frac{\sqrt{3}}{3}=\frac{4\sqrt{3}}{3}$$

3б) Вычислить площадь фигуры, ограниченной

$$y = 2x - x^2,\ y = -x$$

Найдём границы интегрирования

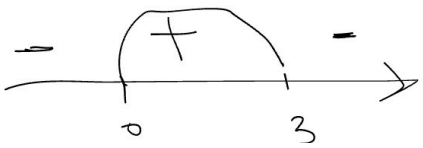
$$2x - x^2 = -x$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$\left[\begin{array}{l} x = 0 \\ x = 3 \end{array}\right.$$

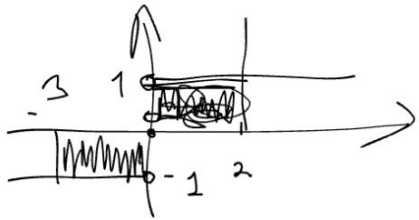
$\left(f(x) - g(x)\right)dx$, если $f(x) \geq g(x)$ на $[a, b]$



$$S = \int_0^3 \left(2x - x^2 - (-x)\right)dx = \int_0^3 \left(3x - x^2\right)dx = \left(\frac{3x^2}{2} - \frac{x^3}{3}\right) \Big|_0^3 =$$

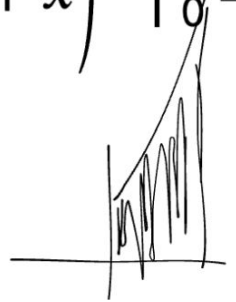
$$= \left(\frac{27}{2} - 9\right) - (0 - 0) = 13.5 - 9 = 4.5$$

4а) $$\int_{-3}^{2} sign(x)dx = \int_{-3}^{0} -1dx + \int_{0}^{2} 1dx = -1$$



4б) $$\int_{0}^{5} \frac{3x^2-3}{x-1}dx = 3\int_{0}^{5} \frac{x^2-1}{x-1}dx = 3\int_{0}^{5} (x+1)dx = 3\left(\frac{x^2}{2}+x\right) \Big|_0^5 =$$

$$= 3\left(\frac{25}{2}+5\right) - 3(0+0) = 3 \cdot 17.5 = 52.5$$



5) $\int_1^3 e^{2x}dx = \frac{1}{2}\int_1^3 e^{2x}\cdot 2dx = \frac{1}{2}\int_2^6 e^u\cdot du = \frac{1}{2}e^u \Big|_2^6 = \frac{e^6}{2} - \frac{e^2}{2}$

$u = 2x, \qquad x = 1 \quad x = 3$

$du = 2dx \qquad u = 2 \quad u = 6$

$$\int_1^3 e^{2x}dx = \frac{e^{2x}}{2}\Big|_1^3 = \frac{e^6}{2} - \frac{e^2}{2}$$

6) $$\int sinx \cdot cosx\, dx = \left\{ \begin{array}{c} u = sinx \\ du = cosx\, dx \end{array} \right\} = \int u \cdot du = \frac{u^2}{2} + C = \frac{\left(sin^2 x\right)}{2} + C$$

$$\frac{1}{2} \int 2sinx \cdot cosx\, dx = \frac{1}{2} \int sin(2x)\, dx = \frac{1}{2} \cdot \frac{-cos(2x)}{2} + C = -\frac{1 - 2sin^2 x}{4} + C =$$

$$= -\frac{1}{4} + \frac{sin^2 x}{2} + C$$

7) $$\int tgx\, dx = \int \frac{sinx}{cosx} dx = \left\{ \begin{array}{c} u = cosx \\ du = -sinx\, dx \end{array} \right\} = -\int \frac{-sinx\, dx}{cosx} = -\int \frac{du}{u} =$$

$$= -ln|u| + C = -ln|cosx| + C$$

$$8)\quad \int_0^2 x^2 e^{x^3} dx = \left\{ \begin{array}{c} u = x^3 \\ du = 3x^2 dx \\ x = 0,\ x = 2 \\ u = 0,\ u = 8 \end{array} \right\} = \frac{1}{3} \int_0^2 e^{x^3} \cdot 3x^2 dx = \frac{1}{3} \int_0^8 e^u \cdot du = \frac{e^u}{3} \Big|_0^8 =$$

$$= \frac{e^8}{3} - \frac{e^0}{3} = \frac{e^8}{3} - \frac{1}{3}$$

На сегодня всё!
Вопросики?